Generic Identifiability and Directed Containment for Strongly Tree-Child Level-2 Networks under the Kimura Two-Parameter Model

The Principal Positive Domain and Strict Continuous Time

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Abstract

We classify regular full-dimensional stochastic containment among binary standard semi-directed strongly tree-child level-2 phylogenetic networks under the Kimura two-parameter (K2P) model. On the principal positive Fourier domain

$$\mathcal{D}_+ = \{(s, g) : 0 < s < 1,\ 0 < g < 1,\ g > 2s - 1\},$$

a directed containment germ exists if and only if the two labelled networks are isomorphic after independently redirecting ordinary three-cycle factors. The same condition is equivalent to a common full-dimensional regular germ; in particular, no proper one-sided containment occurs. It follows that the semi-directed topology is generically identifiable modulo ordinary triangle redirection, and that its structural triangle class is exactly reconstructible away from a proper algebraic exceptional set.

The proof combines pointwise displayed-quartet inequalities and exact whole-map $\mathcal{T}_i$ identities, an exact two-sector bridge-fibre theorem, physical marginal submersions, a finite localization argument, and a bounded graph-to-algebra classification of cycle and theta factors. The bounded classification is computer-assisted: every directed primitive relation, rank exclusion, restoration parent, transport, and one-/two-port probe is represented by an exact certificate and has a separately implemented replay and targeted mutation tests. We state this residue explicitly rather than replacing it by an unproved hand-orbit claim. The classification transfers to the strict continuous-time domain $0 < s < 1,\ s^2 < g < 1$. Finally, for every $n \geq 3$ we give two weakly but not strongly tree-child level-2 networks whose continuous-time K2P images share a regular germ of dimension $4n - 3$, proving that strong tree-childness is a sharp hypothesis.

Keywords. phylogenetic networks; Kimura two-parameter model; generic identifiability; directed containment; level-2 networks; strongly tree-child networks; algebraic statistics; computer-assisted proof.

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1 Introduction

Phylogenetic networks model reticulate evolutionary events such as hybridization, recombination, and lateral transfer. A network-based Markov model assigns a site-pattern distribution to each choice of topology, edge parameters, and inheritance probabilities. The basic information question is whether different topologies can generate the same distributions, or more strongly, whether one open model image can contain a full-dimensional regular piece of another.

The Kimura two-parameter model was introduced to distinguish transition and transversion rates while retaining a uniform stationary distribution [13]. Its group-based Fourier form follows the general framework of Evans and Speed and of Sturmfels and Sullivant [6, 16]. Algebraic invariants and Jacobian or algebraic-matroid methods have separated important network classes, including single-cycle and level-1 networks [7, 9, 8]. Recent work proves full K2P identifiability for level-1 networks modulo triangle redirection [3, Theorem 4.9]. Three-sunlet group-based varieties have also been studied directly [4]. At level two, invariant calculations for simple and semisimple networks and quartet-based reconstruction for outer-labelled planar galled networks give complementary partial classifications [1, 10].

Huber et al. [12, Lemma 4.2 and Figure 8] classify the two semi-directed level-2 generator types underlying simple strict level-2 networks. Englander et al. [5] use these generators in their strongly tree-child analysis. Their Propositions 2.9–2.10 and Theorem 2.11 prove, for JC and K2P, pointwise separation of networks with different displayed-quartet sets; their Corollary 2.12 combines this separation with the quarnet structure to recover the labelled tree of blobs. Their main level-2 algebraic classification is for JC.

The directed ported refinement used here, and also in the companion JC classification [14], additionally records the incoming source, internal reticulation events, path-sink child roles, minimum strong repairs, and ordered boundary words. This refinement produces the four event placements $\theta_0, \ldots, \theta_3$. The present work proves that graph layer in proposition 6.2 and rebuilds every algebraic layer for K2P, whose nonzero Fourier characters split into two orbits rather than one.

Our classification concerns a source-relative analytic containment with a physical target section, not merely inclusion of complex varieties. This distinction is important. Boundary degenerations, lower-dimensional intersections, and equality of Zariski closures do not by themselves create an open stochastic containment. Conversely, an open containment cannot be excluded only by comparing generic dimensions when the source can enter an exceptional target locus. The pointwise topology inequalities and the physical localization theorem below address these issues directly.

The proof has five layers.

- **L1.** Exact quartet and three-leaf inequalities recover the labelled decorated tree of blobs at every strict physical point.
- **L2.** Positive rank-one bridge factorizations yield precisely two incidence scales per component–bridge incidence: one for the $\{C, T\}$ sector and one for the G sector. Explicit analytic normalizers and strict serial subdivisions turn the ambient quotient into a physical local product chart.
- **L3.** Marginal maps are physical submersions. A finite semialgebraic-cover argument localizes a global directed relation to one incoming and completion type of each primitive factor; remote components cannot cancel a local certificate.
- **L4.** A bounded exact computation classifies every primitive four- and five-port direction and discharges every fixed-full restoration obligation. Separately certified one- and two-port probes recover arbitrary labelled subdivision words.

L5. Ordinary triangle orientations share a strict rank-nine K2P germ. Contextual contraction and simultaneous physical bridge gluing give the converse global relation.

The computer-assisted fourth layer is finite and exact, but it remains a load-bearing residue. Its theorem object records both graphs, direction, incoming roles, port permutations, omitted roles, transports, polynomial or rank certificates, and parent–child provenance. The reader supplement gives the census, proof schema, representative examples, replay commands, and a claim-to-artifact crosswalk. No claim that the millions of raw presentations have already been reduced to a complete short hand proof is made here.

2 Networks, K2P, and observational relations

2.1 Rooted and semi-directed conventions

Definition 2.1 (Rooted binary network). *Let $\mathcal{X}$ be a finite set of leaf labels. A rooted binary phylogenetic network on $\mathcal{X}$ is a finite directed acyclic graph without parallel arcs in which the root has bidegree $(0,2)$, each tree vertex has bidegree $(1,2)$, each reticulation has bidegree $(2,1)$, and each leaf has bidegree $(1,0)$ and a unique label in $\mathcal{X}$. Every vertex is reachable from the root. We require the root to be the lowest stable ancestor of the labelled leaves: it lies on every root-to-leaf path, and no proper descendant has that property. The presentation is* tree-child *if every internal vertex has a tree vertex or leaf as a child.*

Definition 2.2 (Standard semi-directed network). *From a rooted binary network, mark every arc entering a reticulation, undirect every other arc, delete the binary root, and merge its two incident arcs into one edge. At either endpoint retain exactly the arrowhead present before root deletion. A presentation is admissible here only when this one suppression produces a simple binary mixed graph: it creates neither a loop nor a parallel edge and loses no reticulation arrowhead. No later exhaustive cleanup is part of this operation. The resulting labelled mixed graph is a* standard semi-directed network. *An* admissible rooting *of it is a rooted binary network whose one-step suppression is exactly that mixed graph.*

Isomorphisms preserve leaf labels, ordinary edges, retained arrowheads, and vertex roles. This fixed-mixed-graph convention is the one used by the primitive compiler. Restriction to a leaf subset is a different operation: after choosing displayed reticulation parents, one may prune unlabelled dead ends and suppress unlabelled bivalent vertices. Such cleanup does not enlarge the admissible-rooting set of the original mixed graph. Standard references on rooted and semi-directed conventions include [15, 11].

Definition 2.3 (Weak and strong tree-childness). *A standard semi-directed network is* weakly tree-child *when at least one admissible rooting is tree-child. It is* strongly tree-child *when it has an admissible rooting and every admissible rooting is tree-child.*

Lemma 2.4 (No-omnian criterion). *In the binary fixed convention, a standard semi-directed network with an admissible rooting is strongly tree-child if and only if every tail of a retained reticulation edge is incident with two ordinary edges.*

Proof. If a tail fails the condition, binary degree leaves two cases. It is a reticulation with a reticulation child, in which case every admissible rooting already fails tree-childness; or it is an ordinary vertex tailing two retained edges. A rooting away from those two edges makes both reticulations children of that vertex. If an admissible rooting instead subdivides one retained edge, move the root to the tail's unique ordinary incidence. The old root suppresses back to the first retained edge, all other arrowheads are fixed, and acyclicity is preserved because the tail had indegree one. The resulting admissible rooting has two reticulation children at the tail and is not tree-child. Thus failure of the incidence condition prevents strong tree-childness.

Conversely, assume the condition. In any admissible rooting, an ordinary vertex with one reticulation child has, besides that retained edge, two ordinary incidences; one is its parent and the other is a tree or leaf child. A reticulation cannot tail a retained edge, since its two incoming retained edges plus such an outgoing retained edge would leave no two ordinary incidences. Hence it has an ordinary child. A root inserted on an ordinary edge has a tree or leaf endpoint, while a root inserted on an edge entering a reticulation has the ordinary tail as a tree child. A root with two reticulation children would suppress to an edge with two arrowheads, excluded by definition 2.2. Every admissible rooting is therefore tree-child. □

This is the local criterion used in the primitive repair table of section 6.

A *bridge* is an edge whose deletion disconnects the underlying graph. A *blob* is a maximal nontrivial biconnected block. A network is *level two* when every blob contains at most two reticulations. Contract each blob and retain the intervening ordinary branching components; the resulting labelled tree is the *tree of blobs*. Cutting a blob from the network turns every incident bridge into a labelled boundary *port*. A *complete ported factor* retains every physical port, including a child port below each path-sink reticulation.

Definition 2.5 (Ordinary triangle relation)**.** *An ordinary triangle redirection changes only which vertex of a three-cycle factor incident with exactly three external arms is reticulate and redirects the two arrowheads entering it. Write $N \equiv_\triangle N'$ when the labelled decorated trees of blobs agree, corresponding nontriangle factors are labelled mixed-graph isomorphic, and each remaining corresponding factor differs by such a redirection, with coherent boundary transports.*

This is a structural quotient. It does not assert equality of the complete stochastic images of two redirected networks.

2.2 The K2P network map

Identify the four states and their characters with $\mathbb{Z}_2 \times \mathbb{Z}_2 = \{0, C, G, T\}$, where addition is the Klein-four-group operation. A K2P edge e has Fourier spectrum

$$\widehat{f}_e = (1, s_e, g_e, s_e).$$

Equivalently, if $M_e(x, y) = f_e(y - x)$, inverse Fourier transformation gives

$$f_e(0) = \frac{1 + 2s_e + g_e}{4}, \quad f_e(C) = f_e(T) = \frac{1 - g_e}{4}, \quad f_e(G) = \frac{1 - 2s_e + g_e}{4}. \tag{1}$$

The root state is uniform. At each reticulation r, one incoming edge is selected with probability $\lambda_r \in (0, 1)$ and the other with probability $1 - \lambda_r$. A vector of choices σ gives a displayed tree T_σ, with positive weight w_σ.

For a character assignment $\mathbf{h} = (h_i)_{i \in \mathcal{X}}$, the Fourier network parameterization is

$$q_N(\mathbf{h}) = \begin{cases} \sum_\sigma w_\sigma \prod_{e \in T_\sigma} \widehat{f}_e(\oplus_{i \in A_e} h_i), & \oplus_{i \in \mathcal{X}} h_i = 0, \\ 0, & \text{otherwise}, \end{cases} \tag{2}$$

where $A_e \mid A_e^c$ is the displayed-tree split. This is a polynomial map in the Fourier edge coordinates and inheritance probabilities.

Strict positivity of all four entries in (1) is equivalent to

$$1 - g > 0, \qquad 1 + 2s + g > 0, \qquad 1 - 2s + g > 0.$$

The principal positive-eigenvalue component is

$$\mathcal{D}_+ = \{(s, g) : 0 < s < 1,\ 0 < g < 1,\ g > 2s - 1\}. \tag{3}$$

Every edge in the main theorem lies in $\mathcal{D}_+$, and every inheritance probability lies in $(0, 1)$. Write $\Theta_+(N)$ for this open parameter space, Φ_N for (2), and $\mathcal{M}_+(N) = \Phi_N(\Theta_+(N))$. Put

$$d_N = \max_{\theta \in \Theta_+(N)} \operatorname{rank} D\Phi_N(\theta).$$

A source parameter is *regular* if its Jacobian rank is d_N.

2.3 Directed containment and common germs

Definition 2.6 (Regular analytic relations)**.** *Write $N \preceq_+ N'$ if there are a regular source point $\theta_0 \in \Theta_+(N)$, a connected open neighborhood $U \subseteq \Theta_+(N)$ on which Φ_N has rank d_N, and a real analytic physical target section $\sigma : U \to \Theta_+(N')$ satisfying*

$$\Phi_N = \Phi_{N'} \circ \sigma \qquad \text{on } U.$$

Write $N \bowtie_+ N'$ if the two images contain a common analytic germ which is regular and full-dimensional in both images and which has physical analytic sections from both parameter spaces.

These relations are deliberately stronger than nonempty image intersection. They are distinct from equality of complete stochastic images, inclusion or equality of complex Zariski closures, lower-dimensional coincidence, and boundary-only intersection. The main theorem proves that $\preceq_+$ is symmetric inside the stated class; symmetry is not built into the definition.

3 Physical domain and pointwise topology recovery

Lemma 3.1 (Strict subdivision)**.** *Every $(s, g) \in \mathcal{D}_+$ factors into two strict pairs in $\mathcal{D}_+$. More precisely, for all sufficiently small $\varepsilon > 0$,*

$$(s_B, g_B) = (1 - \varepsilon, 1 - \varepsilon), \qquad (s_A, g_A) = \left(\frac{s}{1 - \varepsilon}, \frac{g}{1 - \varepsilon}\right)$$

are in $\mathcal{D}_+$ and multiply coordinatewise to (s, g).

Proof. At $\varepsilon = 0$, the second pair is the given interior point and the first approaches the identity from within $\mathcal{D}_+$. All defining inequalities are strict and continuous. Hence they persist on a sufficiently small positive interval. This avoids the false general assertion that coordinatewise square roots of every strict stochastic K2P edge are physical. □

Lemma 3.2 (Root movement)**.** *For symmetric K2P, the map Φ_N is unchanged by moving the root to any admissible rooting of the same standard semi-directed graph.*

Proof. Reverse only ordinary arcs on an old-to-new root path. K2P transition matrices are symmetric and the root distribution is uniform, so every displayed-tree probability is unchanged. Use lemma 3.1 at the new root and suppress the old root. If the old root is adjacent to an edge entering a reticulation, inspect each switching. When that parent is chosen, the two old root arms enter only through their serial products. When the other parent is chosen, the remaining one-child stem subtends every observed leaf, carries total Fourier character zero, and contributes multiplier one. Thus every summand in (2) is unchanged. Strong tree-childness supplies the admissible tree-child rootings used later, but the algebraic invariance is reversibility, subdivision, and switching. ☐

3.1 Whole-map quartet separation

Let the three quartet splits be $A = 12 \mid 34$, $B = 13 \mid 24$, and $C = 14 \mid 23$. Define

$$F_A = q_{CCCC} - q_{CCTT}, \tag{4}$$

$$G_B = q_{CCCC} - q_{CCTT} - q_{CTTC} + q_{CTCT}. \tag{5}$$

Lemma 3.3 (Displayed-quartet sign theorem)**.** *On a strict physical K2P quartet tree, F_A is zero on topology A and strictly positive on B and C. The polynomial G_B is zero on A and C and strictly positive on B. Consequently two strict network restrictions with different displayed-quartet sets have disjoint physical images.*

Proof. The equal K2P character sector in the convention of (1) is $\{C, T\}$. Let $P = s_1 s_2 s_3 s_4 > 0$, and let $g_I \in (0, 1)$ be the singleton-sector eigenvalue on the internal quartet edge. Direct substitution in (2) gives the complete pullback table

	A	B	C
F_A	0	$P(1 - g_I)$	$P(1 - g_I)$
G_B	0	$2P(1 - g_I)$	$0.$

Thus the asserted zeros and strict signs hold on all of $\mathcal{D}_+$. Leaf permutations give F_t, G_t for every split t; the global $C \leftrightarrow T$ symmetry gives the equivalent character transport. These are the current-paper coordinates corresponding to the equal-sector formulas in Englander et al.

A network tensor is a positive mixture of all displayed-tree tensors. If one displayed set is the singleton $\{t\}$ and the other contains a different topology, F_t is zero on the first map and positive on the second. If both displayed sets have size at least two and differ, choose a topology t present in one and absent from the other; the corresponding G_t is positive on the first mixture and zero on the second. These cases exhaust the nonempty subsets of the three quartet topologies. This is the explicit K2P specialization of the pointwise theorem of Englander et al. [5, Propositions 2.9–2.10 and Theorem 2.11]. ☐

Marginalization sets omitted Fourier characters to zero. Equality of full tensors would therefore imply equality of every quartet marginal. By [5, Corollary 2.12], lemma 3.3 gives:

Corollary 3.4 (Tree-of-blobs recovery)**.** *Every tensor in the strict image determines the labelled tree of blobs pointwise. In particular, $N \preceq_+ N'$ forces the same labelled tree of blobs.*

3.2 The whole-map $\mathcal{T}_i$ polynomial

For a three-leaf tensor use character zero 0 and put

$$X_h = q_{hh0}, \qquad Y_h = q_{h0h}, \qquad Z_h = q_{0hh} \quad (h = s, g), \qquad V = q_{CTG},$$

where $h = s$ means either member of the equal $\{C, T\}$ sector and $h = g$ means G. Define

$$\mathcal{T}_3 = V^2 X_g - X_s^2 Y_g Z_g. \tag{6}$$

Lemma 3.5 (Whole-map tree–sunlet strict sign)**.** *The polynomial $\mathcal{T}_3$ vanishes on every three-leaf K2P tree. For the three-sunlet whose reticulation is incident with leaf three, write its Fourier map as*

$$q_{xyz} = a_x b_y c_z\{\delta f_y d_z + (1-\delta) f_x e_z\}, \qquad x + y + z = 0, \tag{7}$$

with edge spectra a, b, c, d, e, f and $0 < \delta < 1$. Then

$$\mathcal{T}_3 = -a_s^2 b_s^2 a_g b_g c_g^2 f_s^2 \delta(1-\delta) d_g e_g (1-f_g)^2 < 0. \tag{8}$$

The other two orientations have the corresponding leaf-permuted certificates.

This lemma is an identity on the original full Fourier maps. It is not a classifier obtained by rooting and pruning a three-leaf mixed-graph restriction; that historically tempting shortcut is false for some restored and probe presentations.

Proof. For a three-leaf tree with pendant spectra a, b, c,

$$X_s = a_s b_s, \quad X_g = a_g b_g, \quad Y_g = a_g c_g, \quad Z_g = b_g c_g, \quad V = a_s b_s c_g,$$

so the two monomials in (6) agree. For the sunlet, substitute (7) into (6). After removing the common positive monomial, the remaining identity is

$$f_g\{\delta d_g + (1-\delta)e_g\}^2 - \{\delta d_g + (1-\delta) f_g e_g\}\{\delta f_g d_g + (1-\delta)e_g\} = -\delta(1-\delta) d_g e_g (1-f_g)^2.$$

This gives (8). Every factor after the leading minus sign is strictly positive on $\mathcal{D}_+$. $\square$

Thus the ordinary-vertex/three-sunlet decoration of each degree-three node is pointwise determined. A strongly tree-child theta factor has at least four physical boundaries, so it cannot be confused with an ordinary three-boundary factor. Together with corollary 3.4, this recovers the complete labelled decorated tree of blobs before any finite algebraic classification.

4 The complete two-sector bridge fibre

Cut every bridge of the common decorated tree of blobs. Its component-incidence graph is a tree T. For a component v, let P_v be its normalized Fourier boundary tensor, including retained physical blocks and one character h_e for each incident bridge; thus $P_v(0) = 1$. For a bridge $e = uv$, put

$$k_e(h) = \begin{cases} 1, & h = 0, \\ s_e, & h \in \{C, T\}, \\ g_e, & h = G. \end{cases}$$

Character conservation makes the global contraction

$$Q(\mathbf{h}) = \prod_{v \in V(T)} P_v(\mathbf{h}|_v) \prod_{e \in E(T)} k_e(h_e), \tag{9}$$

where h_e is the group sum of the leaf characters on either side of the bridge cut.

Call a component *marked* when its local tensor retains at least one physical boundary block that can carry a compensating sector character in an anchor entry; otherwise call it *unmarked*. Unmarked retained components have bridge degree at least three.

Lemma 4.1 (Paired-sector uniqueness). *Let a positive normalized component tensor and its incidence-scaled image both be invariant under $C \leftrightarrow T$. If the scale at incidence e is $c_e(h)$, then $c_e(C) = c_e(T)$ for every retained component.*

Proof. Normalize $c_e(0) = 1$ and write $\rho_e = c_e(C)/c_e(T) > 0$. If the component has a retained physical block, compare conservation-supported positive entries carrying C, C, and then T, T, at that block and incidence e, with zero elsewhere. Invariance before and after scaling gives $\rho_e = 1$.

For an unmarked degree-d component, compare the analogous entries at each pair $e \neq f$. This gives $\rho_e \rho_f = 1$. Three distinct incidences and positivity imply every $\rho_e = 1$. The only residual at degree two would be $(\rho_1, \rho_2) = (t, t^{-1})$. No such unmarked factor is retained: cycle, theta, and ordinary components have at least three physical incidences in the strong class, while the only simple two-boundary theta $K_4 - e$ has no tree-child rooting. □

Theorem 4.2 (Complete positive bridge fibre). *On the positive normalized K2P component-tensor locus, the complete fibre of (9) is exactly*

$$P_v \longmapsto P_v \prod_{e \ni v} a_{v,e}^{\mathbf{1}[h_e \in \{C,T\}]} b_{v,e}^{\mathbf{1}[h_e = G]}, \tag{10}$$

$$s_e \longmapsto \frac{s_e}{a_{u,e} a_{v,e}}, \qquad g_e \longmapsto \frac{g_e}{b_{u,e} b_{v,e}}, \tag{11}$$

with positive incidence scales $a_{v,e}, b_{v,e}$. There is no additional positive continuous or discrete gauge, the action is free on every retained component, and the bridge tree has no two-sector holonomy. On physical networks, the fibre is the intersection of this orbit with all local, inheritance, arm, bridge, and K2P inequalities.

Proof. Cut one bridge and fix its character. The corresponding flattening block is a positive rank-one matrix. Positive rank-one factorization is unique up to one positive scalar. All-zero normalization fixes the zero-character scalar; lemma 4.1 identifies the C and T scalars, while the G scalar remains independent. Root T temporarily and peel its leaf components. In the paired and singleton sectors separately, the resulting cut scalars assign one positive scale to every component incidence and give exactly (10)–(11). Induction reaches every component, so no vertex-coupled scale or holonomy remains. Direct cancellation proves the converse, and the observable sector labels exclude a discrete interchange.

It remains to prove freeness and analytic normalization. A marked component uses, for each incidence, a positive conservation-supported entry carrying the chosen sector character at that incidence and at one retained physical block; its exponent matrix is the identity. For an unmarked component of degree $d \geq 3$, use in either sector the pair anchors

$$(1, 2), (1, 3), (2, 3), (1, 4), \ldots, (1, d).$$

Their exponent matrix has leading determinant -2 and rank d. If the observed anchor ratios are r_{ij}, the unique positive scales are

$$a_1 = \sqrt{\frac{r_{12}r_{13}}{r_{23}}}, \quad a_2 = \sqrt{\frac{r_{12}r_{23}}{r_{13}}}, \quad a_3 = \sqrt{\frac{r_{13}r_{23}}{r_{12}}}, \quad a_k = \frac{r_{1k}}{a_1}. \tag{12}$$

Apply the same formula independently in the singleton sector. Positivity makes the normalizers real analytic. The excluded degree-two stabilizer is the only possible failure of freeness. $\square$

The normalizers first produce an ambient positive-tensor quotient. The next lemma supplies the physically essential saturation.

Lemma 4.3 (Physical local product chart). *Near every regular physical point, the quotient in theorem 4.2 is a physical local product chart.*

Proof. Split every effective bridge pair (s, g) into three serial pairs

$$(\alpha_u, \beta_u), \qquad \left(\frac{s}{\alpha_u\alpha_v}, \frac{g}{\beta_u\beta_v}\right), \qquad (\alpha_v, \beta_v).$$

Choose both endpoint pairs sufficiently near $(1, 1)$ that all three lie strictly in $\mathcal{D}_+$, using lemma 3.1 twice. Strictness is open; hence the four endpoint coordinates vary independently in a neighborhood while the residual pair remains physical. Absorbing the endpoint factors into the adjacent tensors realizes all four incidence-scale directions. Perform this independently on the bridge tree and intersect the ambient quotient with a constant-rank physical parameter chart. $\square$

For later use we now make the localized relation precise. If H is a complete ported factor, let $\Theta_+(H)$ contain its internal and retained arm K2P pairs and its inheritance parameters, and let $P_H(\theta)$, with $P_H(0) = 1$, be its normalized boundary Fourier tensor. The positive incidence torus $A_H = (\mathbb{R}^2_{>0})^{\mathrm{ports}(H)}$ acts by

$$P_H(\mathbf{h}) \longmapsto P_H(\mathbf{h}) \prod_e a_e^{\mathbf{1}[h_e \in \{C,T\}]} b_e^{\mathbf{1}[h_e = G]}.$$

On any positive anchor chart, (12) gives an analytic slice for this action. Write $\overline{\Phi}_H$ for the resulting slice map and

$$d_H = \max_{\theta \in \Theta_+(H)} \operatorname{rank} D\overline{\Phi}_H(\theta).$$

After fixing a physical port bijection, define $H \preceq_+ H'$ exactly as in definition 2.6, with $\Phi_N, \Phi_{N'}$ replaced by $\overline{\Phi}_H, \overline{\Phi}_{H'}$, a source neighborhood of rank d_H, and a physical analytic target section. This definition is independent of the anchor chart: two such slices differ only by the free analytic incidence action of theorem 4.2. The notation $H \equiv_\triangle H'$ means the restriction of definition 2.5 to the two complete ported factors with that same port bijection. Thus every local relation used below is a relation between quotient model germs, not merely a comparison of abstract graphs.

5 Marginals, localization, and no compensation

Lemma 5.1 (Paired marginal submersion). *For every $m \geq 1$, the serial-edge map*

$$((s_1, g_1), \ldots, (s_m, g_m)) \longmapsto \left(\prod_i s_i, \prod_i g_i\right)$$

maps $\mathcal{D}_+^m$ onto $\mathcal{D}_+$, has differential rank two, and has a local physical analytic section.

Proof. First the product of physical factors remains physical. It is enough to check two factors. Put $S = s_1 s_2$ and $G = g_1 g_2$. Positivity and the upper bounds are immediate. If $S \le 1/2$, then $G > 2S - 1$ is automatic. If $S > 1/2$, then both $s_i > 1/2$, and

$$G > (2s_1 - 1)(2s_2 - 1) = (2S - 1) + 2(1 - s_1)(1 - s_2) > 2S - 1.$$

Induction proves that the displayed product map lands in $\mathcal{D}_+$.

The case $m = 1$ is immediate. Given $(S, G) \in \mathcal{D}_+$ and $m \ge 2$, choose $r \in (0, 1)$ satisfying

$$\max\{S, G, 2S - G, 0\} < r^{m-1} < 1.$$

Use (r, r) on the first $m - 1$ edges and $(S/r^{m-1}, G/r^{m-1})$ on the last. The displayed inequalities put every factor in $\mathcal{D}_+$, and the same fixed r works on a neighborhood of (S, G). The s- and g-rows of the differential have disjoint nonzero supports and are independent. ☐

Proposition 5.2 (Complete graph-derived marginal descriptor)**.** *Let Y be the retained boundary labels in any selected restriction used below, and extend $\mathbf{x} \in \mathbb{Z}_2 \times \mathbb{Z}_2^Y$ by zero on the omitted labels. For an edge e and a displayed switching σ, let $h_{e,\sigma}(\mathbf{x})$ be the group sum below e when $e \in T_\sigma$, and put $h_{e,\sigma}(\mathbf{x}) = \perp$ otherwise. Define*

$$\epsilon_e^s(\sigma, \mathbf{x}) = \mathbf{1}[h_{e,\sigma}(\mathbf{x}) \in \{C, T\}], \qquad \epsilon_e^g(\sigma, \mathbf{x}) = \mathbf{1}[h_{e,\sigma}(\mathbf{x}) = G].$$

The complete visible signature *of e is the pair of these arrays over all $(\sigma, \mathbf{x})$. Edges with the same non-identically-zero signature form one effective product class E, with coordinates*

$$S_E = \prod_{e \in E} s_e, \qquad G_E = \prod_{e \in E} g_e.$$

Edges with identically zero signatures, and tensor-invisible inheritance coordinates, are forgotten. The exact graph transport permutes surviving inheritance coordinates; only when it reverses the recorded order of the two parents at a retained reticulation does it apply $\lambda_r \mapsto 1 - \lambda_r$. Consequently the effective restriction descriptor is a product of the paired serial maps of lemma 5.1, coordinate projections and permutations, and these recorded affine parent-order flips. It maps the physical parameter domain onto the effective child domain, is a submersion, and has local physical analytic sections.

Proof. Here a reticulation is *tensor-invisible* precisely when, after every choice at the other reticulations is fixed, its two parent choices induce the same retained switching and the same visible edge-sector exponent arrays for every retained character assignment. Only under this condition is its inheritance coordinate forgotten; otherwise the compiler retains it. Set omitted characters to zero in (2) and group full switchings by their induced switching on the retained graph. Equal visible signatures contribute in the same K2P sector in every monomial, hence only through (S_E, G_E). Each invisible reticulation contributes $\lambda + (1 - \lambda) = 1$; every surviving reticulation is carried by the exact graph transport, with a complement precisely when that transport reverses its stored parent order. This gives the asserted factorization. Lemma 5.1 handles each product class, coordinate projections are submersions onto the retained coordinates, and permutations and parent flips are analytic diffeomorphisms of the open physical domain. The flips here are not an inheritance-complement quotient between records: they occur only when the certified graph transport reverses a parent order. ☐

We use the following standard semialgebraic fact in a form that makes the finite-choice step explicit.

Lemma 5.3 (Finite semialgebraic covers). *Let M be a smooth d-dimensional semialgebraic manifold and $S \subseteq M$ semialgebraic. Then $\dim S = d$ if and only if S has nonempty relative interior. Consequently, if a nonempty relatively open set in M is covered by finitely many semialgebraic subsets, one member contains a nonempty relatively open subset.*

Proof. This follows from semialgebraic cell decomposition, invariance of dimension under semialgebraic bijections, the frontier-dimension inequality, and the fact that the dimension of a finite union is the maximum of its dimensions; see [2, Section 2.8]. □

Proposition 5.4 (Physical localization and no compensation). *Suppose $N \preceq_+ N'$. For each pair of corresponding complete ported factors H, H', some fixed target incoming/completion type contains a source-relative full-dimensional regular germ of the projective physical K2P model of H. Parameters in other factors cannot compensate for a local separator.*

Proof. Choose the regular source neighborhood and physical target section from definition 2.6. Shrink around a nonzero maximal source minor. By lemma 4.3, source slice coordinates contain a product box of local factor germs and effective bridge intervals. Fix every coordinate except one focal factor. Positive bridge block factorization and the analytic normalizers extract its projective incidence orbit intrinsically from the global tensor.

Only finitely many target incoming roles and target completion types occur in the primitive grammar. Their semialgebraic realization sets cover the focal source box, because the given target section realizes every point. By lemma 5.3, one type contains a source-open subgerm. The intrinsic bridge extraction does not depend on remote target coordinates, so a polynomial or rank obstruction on the focal orbit cannot be cancelled in a distant component. □

Restoration is used only under one fixed full relation.

Lemma 5.5 (Fixed-full restoration). *Fix actual networks N, N' and a full relation $N \preceq_+ N'$. Fix an omitted physical label, its source insertion edge, its target attachment, and the induced boundary transport. Marginalizing this same relation gives one enumerated child relation, and its source restriction is open. Hence exact separation of every physical child excludes the fixed full parent.*

Proof. Marginalization sets the restored leaf character to zero. The pendant factor becomes one and all newly serial edges multiply to the old effective edge. Proposition 5.2, together with the identity on other source parameters, gives a physical source-open image and retains every inheritance coordinate that is not tensor-invisible. The actual attachments and transport of the fixed networks are one row of the exhaustive child construction. Therefore a separating certificate for every child contradicts the assumed full relation. This argument neither lifts an abstract selected relation nor inverts an arbitrary target deletion map. □

6 Primitive factors and the bounded exact theorem

6.1 Graph reduction

Remove the port arms of one blob and suppress ordinary bivalent vertices that are neither local sources nor reticulation sinks. Record every removed port-bearing subdivision as a labelled ordered word on the resulting directed segment.

Lemma 6.1 (Root reduction to a real incoming port)**.** *Every root-containing strongly tree-child factor has the same open projective K2P germ as an incoming-port presentation at some real leaf-bearing boundary. For a comparison, source and target incoming boundaries may be chosen independently.*

Proof. Starting at an admissible root, repeatedly choose a tree or leaf child. This reaches a labelled leaf without crossing a reticulation. Move the root down that ordinary path to its boundary arm. Reversing the path creates no directed cycle: a first re-entry would give an internal path tree vertex a second parent. The other old root branch reaches a leaf off the path, so no proper descendant of the new root lies on every root-to-leaf path; the new root is again the lowest stable ancestor. One-step suppression returns the same fixed mixed graph, and strong tree-childness makes the new rooting tree-child. Lemma 3.2 preserves the K2P map, while strict subdivision or suppression changes only positive incidence factors on the chosen arm. □

Proposition 6.2 (Cycle–theta core theorem)**.** *Every nontrivial complete blob factor in the strong level-2 class has a cycle core or one of four directed theta cores $\theta_0, \theta_1, \theta_2, \theta_3$. It is recovered uniquely from that core, one labelled ordered port word on each directed segment, and one child port below every path-sink reticulation.*

Proof. Move a root-containing factor to a real incoming boundary using lemma 6.1. Let t, r, v, e be its local tree vertices, reticulations, total internal vertices, and blob edges after deleting port arms. Summing indegrees gives $e = v + r - 1$, and therefore

$$\sum_{w \in V(B)} (\deg_B(w) - 2) = 2e - 2v = 2(r - 1).$$

For $r = 1$, every blob vertex has degree two and the core is a cycle. For $r = 2$, exactly two vertices have degree three and every other vertex has degree two. Biconnectedness makes the blob three internally disjoint paths between the two degree-three poles, hence a theta.

The incoming cut edge is ordinary. Indeed, if an arc $u \to R$ entering a reticulation were a cut edge, the two root-to-parent paths together with the other incoming arc of R would still connect u to R after deleting uR, a contradiction. Its endpoint in the factor is therefore the unique local tree source S. A theta pole already uses all three binary incidences inside the blob, so S lies in the interior of one of the three pole-to-pole paths. An internal reticulation is necessarily a path sink: both path sides enter it and its unique child is its boundary arm.

At most one pole is reticulate. If both poles were reticulate, the path containing S would point from S toward both poles, giving each pole one incoming side. Their remaining bidegrees would force the two event-free paths to point in opposite directions, and those paths would form a directed cycle.

Suppose exactly one pole R is reticulate and the other pole T is a tree vertex. The second reticulation is an internal sink X. If S and X lie on different paths, their paths point respectively away from S and toward X, and the pole bidegrees force the third path from T to R; this is θ_0. If they share a path, its only acyclic order from T to R is

$$T - S - X - R.$$

The order $T - X - S - R$ forces the two event-free paths to have opposite directions and hence creates a directed cycle. Both event-free paths in the valid case point from T to R, giving θ_1.

Suppose instead that both poles are tree vertices. The two reticulations are internal sinks. They cannot lie on one path: the path directions into the two sinks would force S between them, after which the pole bidegrees force the two event-free paths in opposite directions, again producing a directed cycle. Thus the sinks lie on distinct paths. If S lies on the third path, both poles feed both sinks, giving θ_2. Otherwise S shares a path with one sink; choose as first pole the pole whose other two sides point outward. The shared path then has order pole–S–sink–pole and all remaining directions are forced, giving θ_3. Reversing the arbitrary pole names or interchanging the two sinks yields only the displayed symmetries. Hence no source or reticulation-event placement is omitted.

Deleting the incoming arm and suppressing its resulting degree-two source yields one of the two semi-directed level-2 generators classified by Huber et al. [12, Lemma 4.2 and Figure 8]. The four directed incoming-source refinements are exactly the cases above. Reading port vertices along the oriented segments recovers the unique words. □

The segment templates and minimal strong repairs are:

core	directed segments in index order	minimum occupied segments
cycle	$S \to X, S \to X$	$\{0\}, \{1\}$
θ_0	$S \to U, S \to V, U \to X, V \to X, U \to V$	$\{2,3\}, \{3,4\}$
θ_1	$S \to U, S \to X, V \to X, U \to V, U \to V$	$\{2,3\}, \{2,4\}$
θ_2	$S \to U, S \to V, U \to X_0, V \to X_0, U \to X_1, V \to X_1$	$\{2,3\}, \{2,5\}, \{3,4\}, \{4,5\}$
θ_3	$S \to U, S \to X_0, V \to X_0, U \to X_1, V \to X_1, U \to V$	$\{2\}, \{4\}$

Here is a direct proof that the repair table is complete. If an occupied segment ends at a reticulation, its last port-bearing subdivision replaces the original core tail's reticulation incidence by an ordinary edge; the new arrow tail has two ordinary incidences, namely its preceding segment edge and its port arm. Consequently an occupancy set A is valid exactly when it meets every obstruction clause in

core	clauses that A must meet
cycle	$\{0,1\}$
θ_0	$\{3\},\ \{2,4\}$
θ_1	$\{2\},\ \{3,4\}$
θ_2	$\{2,4\},\ \{3,5\}$
θ_3	$\{2,4\}$.

For the cycle, S otherwise tails both reticulation edges and the two unsubdivided sides are parallel. In θ_0, segment 3 must be occupied to separate the reticulation V from its reticulation child X, while U requires one of segments $2, 4$. In θ_1, segment 2 separates the reticulation stack, while one of the parallel segments $3, 4$ must be occupied both to give U an ordinary child and to make the mixed graph simple. In θ_2, U requires one of $2, 4$ and V one of $3, 5$. In θ_3, V requires one of $2, 4$. All other retained-arrow tails already have two ordinary incidences, and every path-sink child arm is retained separately.

The minimal transversals of these clause families are precisely the repair sets displayed in the table. Deleting any member of a displayed repair leaves its named clause unmet, whereas every valid occupancy must meet all the clauses. This proves sufficiency, minimality, and exhaustiveness, up to the displayed core symmetries, by the no-omnian criterion lemma 2.4. The reader supplement records the exact completion grammar.

For a target core H, let m_H be its number of directed segments, q_H its path-sink count, and r_H its number of compiler target-repair tags. For theta targets these tags are the listed minimum repairs;

cycle targets use one untagged target convention. If k physical boundaries are selected and $\epsilon = 1$ when the incoming boundary is selected ($\epsilon = 0$ when it is marginalized), then the exact number of target completions is

$$C(k,\epsilon) = \sum_H r_H \sum_{j=0}^{q_H} \binom{q_H}{j} \binom{k-\epsilon-j+m_H-1}{m_H-1}. \tag{13}$$

This counts repair-tagged directed completion descriptors, not untagged mixed graphs. First choose the j selected path sinks and weakly distribute the remaining $k-\epsilon-j$ selected physical labels among the m_H ordered segment words. For a chosen theta repair, every required segment whose physical word is empty then receives one deterministically named zero-character completion leaf; an already occupied required segment receives no extra leaf. These completion roles consume no selected label and therefore do not alter the stars-and-bars term. The repair index remains part of the raw descriptor even when two repair witnesses construct the same final mixed graph. Unselected path-sink roles and a marginalized incoming role likewise receive deterministic dummy labels. Cycle targets use the single untagged target-repair convention; their two minimum repaired supports enter the source multiplier rather than C.

The five tuples (m_H, q_H, r_H) are

$$(2,1,1),\ (5,1,2),\ (5,1,2),\ (6,2,4),\ (6,2,2)$$

for the cycle and $\theta_0, \ldots, \theta_3$, respectively. Thus $C(4,1) = 831$, $C(4,0) = C(5,1) = 1{,}983$, and $C(5,0) = 4{,}155$. Physical label permutations are applied only after this directed-record enumeration.

6.2 What was computed

For clarity, a *raw directed presentation* consists of the source graph, target graph, source and target incoming modes, all selected or marginalized physical roles, an ordered physical port permutation, the source-to-target direction, and the boundary transport. Canonical classes are formed only after the raw ledger is complete. A row may terminate only by one of the following mutually exclusive evidence types, applied in order:

(i) a pointwise displayed-quartet or whole-map $\mathcal{T}_i$ sign exclusion;
(ii) a directed exact-rank exclusion, with a nonzero source minor and a symbolic target upper certificate;
(iii) an exact polynomial separator, zero on the target map and nonzero at a strict rational source point;
(iv) a fixed-full restoration obligation;
(v) a labelled mixed-graph isomorphism; or
(vi) an ordinary triangle relation.

The target completion grammar takes every primitive core, every allowed path-sink child subset, every weak composition of the available port roles among directed segments, every theta minimum-repair tag (and the single untagged cycle convention), and either a selected incoming boundary or a marginalized incoming role. It produces 831 selected-incoming and 1,983 marginalized-incoming four-port targets. At five ports it produces 1,983 selected-incoming and 4,155 marginalized-incoming targets. These are counts of complete graph-derived target descriptors, not unexplained numerical fit classes.

Theorem 6.3 (Computer-assisted bounded classification)**.** *For every pair H, H' of complete ported physical cycle/theta factors in the strong level-2 class, including arbitrary admissible restored subdivision words,*

$$H \preceq_+ H' \iff H \cong H' \text{ as labelled mixed graphs} \quad \text{or} \quad H \equiv_\triangle H'.$$

Every finite obstruction and every equality transport in this statement has an exact graph-derived certificate. The exhaustive residue consists of the following finite ledgers and forests.

Finite proof and exact residue. For four-port theta supports, six sources, 831 selected-incoming targets, 1,983 marginalized-incoming targets, and 24 physical port permutations give

$$6(831 + 1983)4! = 405{,}216$$

raw directions. The exact partition is

displayed-quartet exclusion	360,408
whole-map $\mathcal{T}_i$ strict-sign exclusion	16,974
directed exact-rank exclusion	23,822
direct-terminal presentation	1,472
restoration-member presentation	2,540
total	405,216

The 1,472 terminal presentations form 934 canonical classes: 839 exact multihomogeneous quadratics, 36 direct polynomial separators (22 quintic, 12 quartic, and 2 cubic), four additional exact hard-case bindings, 20 labelled isomorphisms, and 35 ordinary-triangle relations. The 2,540 restoration members bind to exactly 997 canonical full-parent obligations.

For the four minimum-repaired five-port θ_2 sources,

$$4(1983 + 4155)5! = 2{,}946{,}240$$

directions partition into 2,942,592 quartet exclusions, 2,528 $\mathcal{T}_i$ exclusions, 800 exact-rank exclusions, 240 direct quadratics, and 80 labelled isomorphisms. The 56 dummy-bearing isomorphism roots generate 864 one-parent descendants; their 832 leaves are 760 quartet exclusions and 72 labelled isomorphisms.

For the three-port cycle core, 13,440 base directions partition into 7,452 $\mathcal{T}_i$ exclusions, 5,964 fixed-full restoration obligations, eight labelled isomorphisms, and 16 triangle relations. The 5,964 roots generate 536,364 physical completions: 535,920 quartet exclusions, 300 $\mathcal{T}_i$ exclusions, 132 exact quadratics, and 12 isomorphisms.

The 997 four-port full-parent obligations generate 36,568 first children: 35,758 quartet, 606 $\mathcal{T}_i$, 148 quadratic, 24 transported quartic, and 32 continuations. Those continuations generate 256 second children, 248 quartet and eight $\mathcal{T}_i$. Hence the forest has 36,824 edges, 36,792 separator leaves, maximum depth two, no cycle, no missing child, no unresolved leaf, and coherent transport on every edge. By lemma 5.5, every parent is excluded.

It remains to pass from bounded rigid supports to arbitrary complete words. The probe input has 176 physical equality anchors: 143 isomorphisms and 33 triangles. Every physical mixed edge is an attachment site, including pendant arms, reticulation-incoming edges, and the root-suppressed segment. The exhaustive one-port ledger has 29,964 directions and the two-port ledger has 544,571. Their exact terminal censuses are

$$29{,}964 = 27{,}758 \text{ quartet} + 99\ \mathcal{T}_i + 1{,}915 \text{ isomorphism} + 192 \text{ triangle},$$
$$544{,}571 = 511{,}266 \text{ quartet} + 576\ \mathcal{T}_i + 30{,}969 \text{ isomorphism} + 1{,}760 \text{ triangle}.$$

Every one-port equality has a unique parent transport; every two-port equality restricts coherently to its one-port parent and has its reversed marginal in the ledger. One-port restrictions determine the core segment of each omitted boundary. Two-port restrictions determine the order of every pair on a common segment. Because these comparisons come from actual words, they are transitive and reconstruct a unique total labelled word on each segment. The transported words agree on overlaps; the only remaining choice is the stored coherent orientation of each ordinary triangle.

All stated counts, graph relations, pullbacks, rank bounds, parentage, and transport restrictions were regenerated by an independent implementation. Mutation suites reject an omitted or duplicated raw row, false rank exclusion, missing child, wrong parent, broken transport, and reassignment of quadratic, cubic, quartic, quintic, or $\mathcal{T}_i$ certificates. The supplement identifies the exact files and replay commands. This completes the finite computer-assisted proof; it is not being replaced here by an unverified assertion that all terminal classes form a smaller hand orbit. □

6.3 Bounded proof-compression outcome

A separate bounded normalization and orbit pass was run against the frozen classifier. Its outcome is `PC-PARTIAL`: it substantially reduces the mathematical catalogue, but it does not remove the exact coverage ledgers. The safe reductions are summarized below.

layer	compressed explanation	exact residue retained
primitive completions	one stars-and-bars formula	primitive-core theorem and directed record keys
quadratic separators	literal-body counts (8,4,6,5) for four-port, θ_2, cycle, and restoration layers	every direction-preserving certificate transport
higher-degree direct cases	three certified transport families	27 direction-specific bodies across 36 records
rank exclusions	one universal polynomial-field mechanism	75 exceptional orbit representatives covering 864 descriptors
restoration	297 descriptive outcome fingerprints	all 997 parent assignments and 36,824 labelled forest edges
arbitrary port words	one segment lemma, one order lemma, and coherent triangle transport	29,964 one-port and 544,571 two-port rows

In particular, the 297 restoration fingerprints are not asserted to be a graph-transport quotient, and the three higher-degree propositions are not misreported as three literal polynomials. The compressed verifier checks assignment to these mathematical families and then checks exact agreement with the frozen proof system. This is the stopping point of the one-cycle compression attempt; a further reduction of the retained transport residue would require new mathematics rather than repackaging.

7 Triangle germs and the global classification

7.1 The ordinary-triangle common germ

Let $\tau_i : \Theta_i \to Q$, $i = 1, 2, 3$, be the three normalized K2P maps obtained by placing the reticulation at each vertex of an ordinary three-cycle. Here Q is the nine-dimensional affine space of normalized, conservation- supported three-leaf K2P Fourier tensors.

Lemma 7.1 (Exact rank-nine triangle germ)**.** *The three physical orientation maps have a common strict continuous-time point at which each has rank nine. Consequently they contain one common nine-dimensional regular physical analytic germ.*

Proof. For the orientation at leaf three use (7) and set

$$a = b = c = (1, \tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}), \qquad f = (1, \tfrac{1}{3}, \tfrac{1}{3}, \tfrac{1}{3}), \qquad d = e = (1, \tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}), \qquad \delta = \tfrac{1}{2}.$$

This assignment is invariant under leaf permutations, so all three orientations produce the same tensor. Its six nonconstant two-character orbit coordinates are $1/12$, and its three mixed triple-orbit coordinates are $1/48$. Every edge satisfies $g > s^2$.

At this symmetric point the K2P Jacobian decomposes into four symmetric and five anisotropic output directions. Write $U = q_{CGT}, V = q_{CTG}, W = q_{GCT}$. Use the output logarithms

$$z_0 = (\log X_s, \log Y_s, \log Z_s, \log W),$$

and

$$z_\perp = \big(\log(X_g/X_s), \log(Y_g/Y_s), \log(Z_g/Z_s), \log(U/W), \log(V/W)\big).$$

For the symmetric block vary the common logarithmic s, g scales of a, b, c, f. For the anisotropic block vary $\log(a_g/a_s), \log(b_g/b_s), \log(c_g/c_s), \log(d_g/d_s), \log(f_g/f_s)$, holding the other paired coordinates and the e-anisotropy fixed. Direct differentiation of (7) gives a block-upper-triangular nine-dimensional slice with diagonal blocks

$$J_0 = \begin{pmatrix} 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1/4 \\ 0 & 1 & 1 & 1/4 \\ 1 & 1 & 1 & 1 \end{pmatrix}, \qquad \det J_0 = -\frac{1}{2},$$

and

$$J_\perp = \begin{pmatrix} 1 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 3/4 & 1/4 \\ 0 & 1 & 1 & 1/4 & 1/4 \\ -1 & 1 & 0 & 0 & 0 \\ -1 & 0 & 1 & 1/2 & -1/2 \end{pmatrix}, \qquad \det J_\perp = -\frac{1}{4}.$$

Their product is $1/8$, so every orientation has rank nine. Since $\dim Q = 9$, each orientation map is an analytic submersion at the common tensor. The analytic submersion theorem gives a physical analytic section over an output neighborhood for each orientation; strict continuous time is open, so the sections stay inside that cone after shrinking. Intersecting the three output neighborhoods gives the common regular germ. Notice from (8) that it remains strictly separated from the three-leaf tree model. □

Lemma 7.2 (Contextual triangle sufficiency)**.** *If corresponding factors of two networks are isomorphic or differ only by ordinary triangle redirections, with coherent boundary transports, then the global K2P images share a full-dimensional regular physical germ.*

Proof. An isomorphism transports physical parameters. For one triangle, lemma 7.1 gives a common open set $U \subset Q$ and physical sections of every orientation. Let C be an open strict physical parameter chart of the identical exterior context and write its contraction as

$$\mathcal{H} : Q \times C \longrightarrow Y.$$

Choose (q, c) in the nonempty open subset of $U \times C$ on which $D\mathcal{H}$ has generic maximal rank R. Since each $\tau_i \times \mathrm{id}_C$ is a submersion,

$$\operatorname{rank} D\big(\mathcal{H} \circ (\tau_i \times \mathrm{id}_C)\big) = R.$$

By maximality of R and persistence of a nonzero R-minor, shrink to a neighborhood $V \subset U \times C$ on which $\mathcal{H}$ has constant rank R. The analytic constant-rank theorem supplies a local analytic section $\eta : \mathcal{H}(V) \to V$ over this R-dimensional image germ. If $s_i : U \to \Theta_i$ is the triangle section, then

$$y \longmapsto \big(s_i(\mathrm{pr}_Q\, \eta(y)), \mathrm{pr}_C\, \eta(y)\big)$$

is a physical analytic section of $\mathcal{H} \circ (\tau_i \times \mathrm{id}_C)$. Thus the R-dimensional germ $\mathcal{H}(V)$ is the same full-dimensional contextual image germ for every orientation, even if the three arms reconnect inside another level-2 factor. For several triangles use the product of all Q_j, U_j and their product sections, and make the rank choice once; no induction assumes that an already chosen context remains generic.

Finally glue bridge incidence scales physically. If the two local sections have paired and singleton incidence products $A_e^{(k)}, B_e^{(k)} > 0$, $k = 1, 2$, choose

$$0 < s_e < \min\{\tfrac{1}{2}, A_e^{(1)}/2, A_e^{(2)}/2\}, \qquad 0 < g_e < \min\{1, B_e^{(1)}, B_e^{(2)}\}.$$

The original and both gauge-transformed pairs lie in $\mathcal{D}_+$: every relevant s-coordinate is below $1/2$, so the lower K2P inequality is automatic. Shrink the germs uniformly. Incidence factors cancel termwise in (9), and lemma 4.3 supplies the full-rank global product germ. ∎

Theorem 7.3 (K2P-SAME)**.** *Let N, N' be binary standard semi-directed strongly tree-child level-2 networks on the same finite labelled leaf set. Put every edge in $\mathcal{D}_+$ and every inheritance probability in $(0,1)$. Then*

$$\boxed{N \preceq_+ N' \quad \Longleftrightarrow \quad N \equiv_\triangle N' \quad \Longleftrightarrow \quad N \bowtie_+ N'.} \tag{14}$$

In particular, no proper one-sided regular-germ containment exists in this class. The conclusion is not equality of complete stochastic images.

Proof. Suppose first that $N \preceq_+ N'$. The whole-map inequalities in section 3 force equality of the labelled decorated trees of blobs. The complete bridge fibre and physical product chart extract corresponding projective local factors with exactly the two incidence gauges and no holonomy. Proposition 5.4 selects one physical target type for each source-open local germ and prevents remote compensation. Theorem 6.3 then makes every corresponding complete factor isomorphic or ordinary-triangle related. Its one-/two-port transports reconstruct every labelled segment word and make the local relations globally coherent. Thus $N \equiv_\triangle N'$.

Conversely, $N \equiv_\triangle N'$ implies $N \bowtie_+ N'$ by lemma 7.2. A common full-dimensional germ with physical sections gives the directed relation in both directions by definition. ∎

8 Generic structural identifiability

Let

$$\mathcal{V}_N = \overline{\Phi_N(\mathbb{C}^{m_N})}^Z$$

denote the complex Zariski closure of the polynomial K2P map after complexifying its m_N edge and inheritance coordinates. It is irreducible because it is the closure of the image of an irreducible affine space, and $\dim \mathcal{V}_N = d_N$. Indeed, let r be the generic complex Jacobian rank. Every Jacobian minor has real-polynomial coefficients and some r-by-r minor is nonzero. Since $\Theta_+(N)$ contains a nonempty Euclidean-open subset of $\mathbb{R}^{m_N}$, that polynomial cannot vanish throughout $\Theta_+(N)$; hence $d_N \geq r$. The reverse inequality is immediate, while the generic-rank theorem gives $r = \dim \mathcal{V}_N$.

Theorem 8.1 (Generic identifiability modulo triangles)**.** *For each fixed network N in the class, there is a proper Zariski-closed set $E_N \subsetneq \mathcal{V}_N$ such that every exact physical tensor $q \in \mathcal{M}_+(N) \setminus E_N$ determines the labelled standard semi-directed topology of N uniquely modulo $\equiv_\triangle$.*

Proof. For fixed $\mathcal{X}$, there are finitely many labelled binary strong level-2 topologies. Indeed, choose an admissible rooting $\widetilde{N}$; by strong tree-childness it is tree-child. Starting at the root and at each of its r reticulations, repeatedly choose a tree-vertex or leaf child. The resulting $r+1$ tree paths end at distinct leaves: at the first merger of two paths, a tree vertex or leaf would have two parents. Thus $r \leq |\mathcal{X}| - 1$. If t is the number of nonroot tree vertices, equality of total indegree and outdegree gives

$$2 + 2t + r = t + 2r + |\mathcal{X}|, \qquad t = |\mathcal{X}| + r - 2.$$

Hence

$$|V(\widetilde{N})| = 1 + t + r + |\mathcal{X}| = 2|\mathcal{X}| + 2r - 1 \leq 4|\mathcal{X}| - 3.$$

There are therefore only finitely many labelled rooted presentations, and hence only finitely many standard semi-directed topologies. For each competitor $N' \not\equiv_\triangle N$, form the physical output intersection

$$C_{N,N'} = \mathcal{M}_+(N) \cap \mathcal{M}_+(N')$$

and the target incidence correspondence

$$Z_{N'} = \{(q, \theta') : q = \Phi_{N'}(\theta'),\ q \in \mathcal{M}_+(N)\}.$$

These sets and their projections are semialgebraic by Tarski–Seidenberg [2, Theorem 2.2.1]. Let the *total* source rank-drop locus be

$$R_N = \{\theta \in \Theta_+(N) : \operatorname{rank} D\Phi_N(\theta) < d_N\}.$$

Choose a finite Nash stratification $R_N = \bigsqcup_\alpha S_\alpha$ on which every $\Phi_N|_{S_\alpha}$ has constant rank. Since $T_\theta S_\alpha \subseteq T_\theta \Theta_+(N)$,

$$\operatorname{rank} D(\Phi_N|_{S_\alpha})(\theta) \leq \operatorname{rank} D\Phi_N(\theta) \leq d_N - 1.$$

The constant-rank theorem gives $\dim \Phi_N(S_\alpha) \leq d_N - 1$, so $\Phi_N(R_N)$ is semialgebraic of dimension at most $d_N - 1$. Consequently, a d_N-dimensional physical intersection meets the regular source-image locus in dimension d_N; deleting $\Phi_N(R_N)$ cannot lower its dimension.

Choose finite Nash (real-analytic semialgebraic) stratifications of the regular source image, the target parameter space, and $Z_{N'}$, compatible with the relevant rank loci and refined so that every restricted polynomial map and projection has constant rank on each stratum.

If $C_{N,N'}$ had dimension d_N, then lemma 5.3 would give a nonempty relatively open subgerm in a regular d_N-dimensional source-image stratum. Some Nash stratum of $Z_{N'}$ would project onto a smaller such germ with rank d_N; the analytic constant-rank theorem supplies a physical real-analytic target section over that output germ. An analytic constant-rank chart for Φ_N then pulls this section back to an open source-parameter neighborhood, constant along the local source fibres. This is $N \preceq_+ N'$, contradicting theorem 7.3. Hence every competing physical intersection has semialgebraic dimension at most $d_N - 1$. By [2, Proposition 2.8.2], a semialgebraic set and its real Zariski closure have the same dimension; extension of scalars from $\mathbb{R}$ to $\mathbb{C}$ preserves Krull dimension. Its complex Zariski closure inside $\mathcal{V}_N$ therefore has dimension at most $d_N - 1$ and is proper.

Take the finite union of these closures and the singular locus of $\mathcal{V}_N$. For every output polynomial $P(q)$ used by reconstruction—a quartet, $\mathcal{T}_i$, bridge-anchor, or finite-separator polynomial—adjoin $\mathcal{V}_N \cap Z(P)$ only when $P \circ \Phi_N$ is certified nonidentically zero on the intended source stratum; this intersection is then proper. Also adjoin the complex Zariski closure of $\Phi_N(R_N)$, which is proper by the dimension calculation above and the same real-to-complex closure facts. Every added set is therefore a proper closed subset of $\mathcal{V}_N$; a finite union of such sets cannot fill the irreducible $\mathcal{V}_N$. The resulting proper set is E_N. Outside it, every realizing topology is triangle-equivalent to N. □

The quantifier is "for each fixed N, outside a proper exceptional set." This is not pointwise identifiability at every parameter and does not identify individual bridge parameters inside the incidence-gauge fibre.

9 Exact structural reconstruction

Theorem 9.1 (Finite exact reconstruction)**.** *There is a terminating exact procedure which, given $q \in \mathcal{M}_+(N) \setminus E_N$ for some network N in the class, returns the unique structural class $[N]_\triangle$.*

Proof. Here "exact" means that every input coordinate is supplied in a representation supporting exact field operations, polynomial-sign decisions, and real-closed-field quantifier elimination (equivalently, through an exact-real oracle with those operations). This is a termination model, not a bit-complexity or numerical-stability claim.

Apply the following finite procedure.

- **R1.** Fourier-transform the exact pattern tensor.
- **R2.** Evaluate the quartet sign deck and reconstruct the labelled tree of blobs.
- **R3.** Evaluate all leaf-permuted $\mathcal{T}_i$ and decorate every degree-three component.
- **R4.** Factor positive bridge blocks in the paired and singleton sectors and impose the analytic normalizers (12).
- **R5.** Enumerate the finitely many bounded rigid-support candidates compatible with R2–R4. Apply the pointwise and polynomial separator tests to the recovered local tensors, retaining every candidate not excluded.
- **R6.** For each retained candidate, follow only exact fixed-full restoration records and their stored transports; the depth is at most two.
- **R7.** For each resulting support, use the certified one-port segment deck and two-port order deck to reconstruct every labelled subdivision word.
- **R8.** Assemble every coherent labelled mixed-graph candidate and group the finite list into ordinary-

triangle classes. For each such class $\mathcal{C}$, decide the exact semialgebraic feasibility statement

$$q \in \bigcup_{H \in \mathcal{C}} \mathcal{M}_+(H).$$

Since $q \notin E_N$, exactly one class is feasible; return its canonical representative.

Each feasibility query is a finite existential semialgebraic sentence and terminates by quantifier elimination under the exact-input convention. The true class is feasible. If an inequivalent class were also feasible, then q would belong to a competing physical intersection whose Zariski closure is contained in E_N, a contradiction.

All decks and forests are finite, and every loop terminates. The largest bounded restriction in the input contract uses at most nine physical ports, so a direct implementation uses at most $O(|\mathcal{X}|^9)$ bounded restrictions beyond reading and Fourier-transforming the explicit $4^{|\mathcal{X}|}$-entry tensor. No numerical-stability or bit-complexity bound is asserted. □

The output is a structural triangle class. Deciding whether the particular input tensor lies in the complete stochastic image of another redirected representative is a separate exact semialgebraic membership problem and is not claimed here.

10 Strict continuous-time transfer

For positive continuous-time K2P rates the Fourier coordinates satisfy

$$\mathcal{D}_{\mathrm{CT}} = \{(s, g) : 0 < s < 1,\ s^2 < g < 1\}. \tag{15}$$

This is a nonempty Euclidean-open subset of $\mathcal{D}_+$. Define $\Theta_{\mathrm{CT}}(N)$, $\mathcal{M}_{\mathrm{CT}}(N) = \Phi_N(\Theta_{\mathrm{CT}}(N))$, $\preceq_{\mathrm{CT}}$, and $\bowtie_{\mathrm{CT}}$ by replacing every edge domain by $\mathcal{D}_{\mathrm{CT}}$. Because $\mathcal{D}_{\mathrm{CT}}$ is a nonempty open subset and every maximal Jacobian minor is polynomial,

$$d_{\mathrm{CT},N} := \max_{\Theta_{\mathrm{CT}}(N)} \operatorname{rank} D\Phi_N = d_N.$$

Corollary 10.1 (Continuous-time classification)**.** *The equivalence* (14)*, generic identifiability in theorem 8.1, and reconstruction in theorem 9.1 hold with* $+$ *replaced by* CT*.*

Proof. Necessity follows because $\mathcal{D}_{\mathrm{CT}}$ is open inside $\mathcal{D}_+$. Positive coordinate power roots stay in $\mathcal{D}_{\mathrm{CT}}$, giving continuous-time subdivision and the marginal sections. The triangle witness in lemma 7.1 is strict continuous-time.

For simultaneous bridge gluing, let $A^{(k)}, B^{(k)} > 0$, $k = 1, 2$, be the two sections' incidence products and put

$$L = \max\left\{1, \frac{B^{(1)}}{(A^{(1)})^2}, \frac{B^{(2)}}{(A^{(2)})^2}\right\}, \qquad U = \min\{1, B^{(1)}, B^{(2)}\}.$$

Choose explicitly

$$0 < s < \min\{1, A^{(1)}, A^{(2)}, \sqrt{U/L}\}, \qquad Ls^2 < g < U.$$

Then (s, g) and both transformed pairs $(s/A^{(k)}, g/B^{(k)})$ lie in $\mathcal{D}_{\mathrm{CT}}$. All inequalities are strict and persist on a neighborhood. Finally, every nonzero polynomial witness remains nonzero on a dense open part of the continuous-time parameter space, and the semialgebraic arguments restrict unchanged. □

No boundary case $g = s^2$, $s = 0$, $g = 0$, or inheritance probability in $\{0, 1\}$ is included.

11 Sharpness of strong tree-childness

The following construction is independent of the bounded atlas and probe gate.

Theorem 11.1 (Weak-class $4n-3$ ambiguity)**.** *For every integer $n \geq 3$, there are two binary level-2 standard semi-directed networks on the same n-leaf labelled set which are weakly tree-child but not strongly tree-child, are neither labelled-isomorphic nor ordinary-triangle-equivalent, and whose strict continuous-time K2P images contain a common full-dimensional regular analytic germ of dimension $4n-3$.*

Proof. We first give the three-leaf pair. The rooted presentations W, W' have arcs

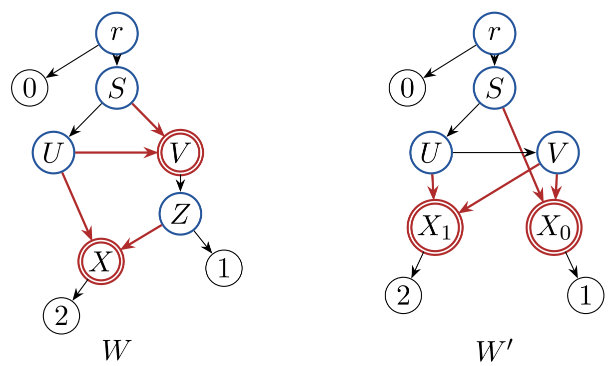
$$\begin{aligned} W:&\quad rS, rL_0, SU, SV, UX, VZ, ZX, UV, ZL_1, XL_2, && \mathrm{Ret}(W) = \{V, X\}, \\ W':&\quad rS, rL_0, SU, SX_0, VX_0, UX_1, VX_1, UV, X_0L_1, X_1L_2, && \mathrm{Ret}(W') = \{X_0, X_1\}. \end{aligned}$$

Suppress r and retain all reticulation arrowheads. Complete edge-rooting enumeration gives

$$(\#\text{ admissible}, \#\text{ tree-child}, \#\text{ non-tree-child}) = (5, 2, 3), \qquad (7, 2, 5).$$

Thus both are weakly but not strongly tree-child. Exact labelled incidence- graph comparison finds no isomorphism. Each underlying graph has one triangle, but forgetting the head flags on those triangle edges still gives no isomorphism, so they are not ordinary-triangle equivalent.

For visual reference, their rooted representatives are:



Put $\delta = 2^{-30}$. In W, assign $(s, g) = (1/7, 1/7)$ to the seven nonpendant arcs

$$rS, SU, SV, UX, VZ, ZX, UV.$$

At reticulation X, give $Z \to X$ weight $15996/16339$ and $U \to X$ its complement $343/16339$. At V, give $S \to V$ weight $1/8$ and $U \to V$ weight $7/8$. Give the pendant arms at leaves $0, 1, 2$ the strict continuous-time diagonal pairs

$$(s, g) = (x_i, x_i), \qquad (x_0, x_1, x_2) = \left(\frac{86779}{80}\delta, \frac{320}{253}\delta, \frac{114373}{20240}\delta\right).$$

In W', assign $(1/4, 1/4)$ to

$$rS, SU, SX_0, VX_0, UX_1, VX_1, UV.$$

At X_1, the edges $V \to X_1, U \to X_1$ have weights $1/2, 1/2$. At X_0, the edges $V \to X_0, S \to X_0$ have weights $1/6, 5/6$. The pendant diagonal pairs at leaves $0, 1, 2$ are

$$\left(\frac{16}{3}\delta, \frac{16}{3}\delta\right), \quad \left(\frac{32}{9}\delta, \frac{32}{9}\delta\right), \quad \left(\frac{96}{5}\delta, \frac{96}{5}\delta\right).$$

Every pair lies strictly in $\mathcal{D}_{\mathrm{CT}}$.

These parameters are written on the displayed rooted arcs. When the artificial root is suppressed, the two incident spectra multiply coordinatewise; the resulting effective semi-directed edge also remains strict continuous-time.

Use numeric character codes $0, C, G, T = 0, 1, 2, 3$. In the explicit coordinate order

$$(000, 0CC, 0GG, C0C, CC0, CGT, CTG, G0G, GCT, GG0), \tag{16}$$

direct four-switch expansion gives the same tensor on both networks:

$$\begin{aligned} q_{000} &= 1, \qquad q_h = \delta^2 && \text{on the six two-nonzero coordinates,} \\ q_h &= \frac{4}{5}\delta^3 && \text{on the three all-nonzero coordinates.} \end{aligned}$$

The graph-derived canonical descendant-signature orders of the seven visible edge classes are

$$\begin{array}{c|ccccccc} W & ZX & SV & rS & SU & UV & VZ & UX \\ W' & VX_1 & VX_0 & UV & rS & SX_0 & SU & UX_1. \end{array}$$

Order each normalized map's full parameter list by the paired (s, g) coordinates of these seven classes, followed by λ_0, λ_1. Thus the named minor columns are

$$\begin{aligned} W &:;(s_{ZX}, g_{ZX}, s_{SV}, g_{SV}, s_{rS}, g_{rS}, s_{SU}, g_{SU}, s_{UV}), \\ W' &:;(s_{VX_1}, g_{VX_1}, s_{VX_0}, g_{VX_0}, s_{UV}, g_{UV}, s_{rS}, g_{rS}, s_{SX_0}). \end{aligned}$$

All output-row indices below are zero-based in the order (16); thus row 0 is q_{000}. The derived column crosswalk in the supplement independently reconstructs these orders from the graph encodings and binds them to the frozen descriptor hashes. For W, rows $(1, 2, 3, 5, 4, 7, 6, 8, 9)$ of (16) and the displayed named columns give determinant

$$\frac{10368019213741323}{5639813150744640239644423884648889156342906888} \neq 0.$$

For W', rows $(1, 2, 3, 5, 4, 6, 7, 8, 9)$ and its displayed named columns give

$$\frac{1435825}{850025966916536138465528} \neq 0.$$

Both normalized maps therefore have rank nine, the full normalized three-leaf dimension. The nonzero pendant factors are an invertible diagonal row scaling, so both full physical maps are submersions at their common tensor. Their images share a regular nine-dimensional germ.

To pass from n leaves to $n+1$, replace the same labelled leaf on both networks by the same labelled cherry. If the two new edge pairs are (u_s, u_g) and (v_s, v_g), call the new leaves a, b, and choose any outside leaf j. On the positive common germ the following Fourier coordinates are nonzero, and character conservation gives

$$R_s = \frac{Q_{j=C, a=C, b=0, \text{ others}=0}}{Q_{j=C, a=0, b=C, \text{ others}=0}} = \frac{u_s}{v_s}, \qquad P_s = Q_{a=C, b=C, \text{ others}=0} = u_s v_s.$$

Replacing C by G gives $R_g = u_g/v_g$ and $P_g = u_g v_g$. Thus the four quantities below are tensor observables, not auxiliary parameter functions:

$$R_s = u_s/v_s, \quad P_s = u_s v_s, \qquad R_g = u_g/v_g, \quad P_g = u_g v_g.$$

Their Jacobian determinant is

$$\det \frac{\partial(R_s, P_s, R_g, P_g)}{\partial(u_s, v_s, u_g, v_g)} = \frac{4u_s u_g}{v_s v_g} \neq 0. \tag{17}$$

For example, the actual pairs $(u_s, u_g) = (2/5, 4/9)$ and $(v_s, v_g) = (3/7, 5/11)$ lie in $\mathcal{D}_{\mathrm{CT}}$, and (17) equals 2464/675. The extended tensor factors through the old tensor and the four new edge coordinates, giving an upper bound of four new dimensions. The inverse formulas $u_s = \sqrt{R_s P_s}$, $v_s = \sqrt{P_s/R_s}$, and their g-sector analogues recover those four coordinates analytically. Every old tensor coordinate is then recovered from a cherry coordinate by dividing by its corresponding nonzero u- or v-factor; the old all-zero coordinate is one. Thus these observables together with the old tensor give a local inverse. Both extended images contain the image of the same analytic map from the common old germ times this four-edge chart. Thus each cherry adds exactly four dimensions, and iteration gives $9 + 4(n - 3) = 4n - 3$.

A tree-child and a non-tree-child base rooting both lift through every cherry. Pruning the newest labelled cherry and suppressing its parent returns the base graph. Hence weak-not-strong status, level two, binary degree, nonisomorphism, and failure of triangle equivalence all persist. □

Thus "strongly" in theorem 7.3 cannot be weakened to "weakly," even in strict continuous time. The theorem does not say that every weak network is ambiguous.

12 Scope, proof status, and reproducibility

The main classification excludes stochastic boundary points, singular edge eigenvalues, inheritance probabilities zero or one, nonbinary networks, networks of level greater than two, merely weakly tree-child networks, and every mixed-sign K2P component outside $\mathcal{D}_+$. It classifies regular analytic containment germs and generic structural recovery, not full-image equality, universal numerical-parameter identifiability, noisy-data estimation, conditioning, or finite-sample model selection.

The finite theorem is supported by a deterministic proof package. A quick entry point checks the frozen manifests, exact certificates, cross-layer censuses, and independent replays. A full entry point additionally regenerates the raw four-port and five-port ledgers, rank certificates, direct polynomial overlay, restoration forest, cycle closure, and one-/two-port probe package from primitive graph encodings. Mutations target every load-bearing failure class listed in the proof of theorem 6.3. The supplement records the exact paths, SHA-256 bindings, environment, and commands. The theorem relies on this exact finite computation; "independent replay" means a separately implemented program, not an independent human review.

Data and code availability

The exact certificate and replay files are contained in the public repository `https://github.com/AlecKriebel/Math` under `k2p_level2_identifiability_closure`. The designated versioned annotated source tag for this submission is `k2p-same-biorxiv-v1.0.4`; no DOI is claimed. Its tag-object and peeled-commit identifiers are recorded in external release metadata after the final source commit is created. The article, supplement, and certificate data are licensed under CC BY 4.0, and the verifier code is licensed under the MIT License. The reader supplement provides a relative-path crosswalk that remains valid in the local reproducibility tree; the deterministic archive hash is emitted as external release metadata to avoid a self-referential archive.

Author contributions

Alec Kriebel conceived and directed the study, fixed the mathematical scope and conventions, adjudicated proof and audit results, curated the exact reproducibility package, and accepts responsibility for the final claims and submission.

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Use of generative AI

Generative AI systems assisted with mathematical exploration, code generation, implementation, drafting, and adversarial review. Several earlier automated claims failed audit, including a reciprocal-only bridge chart and incomplete restoration/probe narratives; those versions are retained only as quarantined history and are not theorem evidence. The active claims are supported by exact graph-derived certificates, direct derivations, separately implemented replays, and fail-closed mutation tests. No independent human specialist review is claimed. The author accepts responsibility for the mathematical statement, code, and final submission.

References

[1] Muhammad Ardiyansyah. Distinguishing level-2 phylogenetic networks using phylogenetic invariants, 2021.

[2] Jacek Bochnak, Michel Coste, and Marie-Françoise Roy. *Real Algebraic Geometry*, volume 36 of *Ergebnisse der Mathematik und ihrer Grenzgebiete*. Springer, Berlin, 1998.

[3] Jari Brits, Niels Holtgrefe, Leo van Iersel, and Samuel Martin. On tree–network distinguishability and full identifiability of phylogenetic networks, 2026. Version 3, 25 August 2026; Theorem 4.9.

[4] Shelby Cox, Elizabeth Gross, and Samuel Martin. Group-based phylogenetic models on 3-sunlet networks. *Bulletin of Mathematical Biology*, 87:132, 2025.

[5] Aviva K. Englander, Martin Frohn, Elizabeth Gross, Niels Holtgrefe, Leo van Iersel, Mark Jones, and Seth Sullivant. Identifiability of phylogenetic level-2 networks under the jukes–cantor model. *bioRxiv*, 2025. bioRxiv preprint, first posted 24 April 2025; version 4 revised 4 July 2026.

[6] Steven N. Evans and Terry P. Speed. Invariants of some probability models used in phylogenetic inference. *The Annals of Statistics*, 21(1):355–377, 1993.

[7] Elizabeth Gross and Colby Long. Distinguishing phylogenetic networks. *SIAM Journal on Applied Algebra and Geometry*, 2(1):72–93, 2018.

[8] Elizabeth Gross, Leo van Iersel, Remie Janssen, Mark Jones, Colby Long, and Yukihiro Murakami. Distinguishing level-1 phylogenetic networks on the basis of data generated by markov processes. *Journal of Mathematical Biology*, 83(3):32, 2021.

[9] Benjamin Hollering and Seth Sullivant. Identifiability in phylogenetics using algebraic matroids. *Journal of Symbolic Computation*, 104:142–158, 2021.

[10] Niels Holtgrefe, Elizabeth S. Allman, Hector Baños, Leo van Iersel, Vincent Moulton, John A. Rhodes, and Kristina Wicke. Distinguishing phylogenetic level-2 networks with quartets and inter-taxon quartet distances. *Bulletin of Mathematical Biology*, 87(12):168, 2025.

[11] Niels Holtgrefe, Katharina T. Huber, Leo van Iersel, Mark Jones, and Vincent Moulton. Characterizing semi-directed phylogenetic networks and their multi-rootable variants. *Theory in Biosciences*, 145:4, 2026.

[12] Katharina T. Huber, Leo van Iersel, Mark Jones, Vincent Moulton, and Leonie Veenema-Nipius. When are quarnets sufficient to reconstruct semi-directed phylogenetic networks? *Bulletin of Mathematical Biology*, 87(10):136, 2025.

[13] Motoo Kimura. A simple method for estimating evolutionary rates of base substitutions through comparative studies of nucleotide sequences. *Journal of Molecular Evolution*, 16(2):111–120, 1980.

[14] Alec Kriebel. Strong tree-childness is a sharp generic-identifiability boundary for level-2 jukes–cantor networks. Zenodo preprint, 2026. Version 1.1.7; certificate and reproducibility dataset DOI: 10.5281/zenodo.22064121.

[15] Charles Semple and Mike Steel. *Phylogenetics*, volume 24 of *Oxford Lecture Series in Mathematics and Its Applications*. Oxford University Press, Oxford, 2003.

[16] Bernd Sturmfels and Seth Sullivant. Toric ideals of phylogenetic invariants. *Journal of Computational Biology*, 12(2):204–228, 2005.